

LECTURE 4: PERMANENT MAGNETIC BLDC MOTOR DRIVES

By using high-energy PMs as the field excitation mechanism, a PM motor drive can be potentially designed with high power density, high speed, and high operation efficiency. These prominent advantages are quite attractive for their application in EVs and HEVs. Of the family of PM motors, the BLDC motor drive is the most promising candidate for EV and HEV applications.

The major advantages of BLDC motor includes the following:

- **High efficiency:** BLDC motors are the most efficient of all electric motors. This is due to the use of PMs for the excitation, which consume no power. The absence of mechanical commutator and brushes means low mechanical friction losses and therefore higher efficiency.
- **Compactness:** The recent introduction of high-energy density magnets (rare-earth magnets) has allowed achieving very high flux densities in the BLDC motor. This allows achieving accordingly high torques, which in turns allows making the motor small and light.
- **Ease of control:** The BLDC motor can be controlled as easily as a DC motor because the control variables are easily accessible and constant throughout the operation of the motor.
- **Ease of cooling:** There is no current circulation in the rotor. Therefore, the rotor of a BLDC motor does not heat up. The only heat production is on the stator, which is easier to cool than the rotor because it is static and on the periphery of the motor.
- **Low maintenance, great longevity, and reliability:** The absence of brushes and mechanical commutators suppresses the need for associated regular maintenance and suppresses the risk of failure associated with these elements. The longevity is therefore only a function of the winding insulation, bearings, and magnet life-length.
- **Low noise emissions:** There is no noise associated with the commutation because it is electronic and not mechanical. The driving converter switching frequency is high enough so that the harmonics are not audible.

However, BLDC motor drives also suffer from some disadvantages as follows:

- **Cost:** Rare-earth magnets are much more expensive than other magnets and result in an increased motor cost.
- **Limited constant power range:** A large constant power range is crucial to achieving high vehicle efficiencies. The PM BLDC motor is incapable of achieving a maximum speed greater than twice the base speed.
- **Safety:** Large rare-earth PMs are dangerous during the construction of the motor because flying metallic objects are attracted towards them. There is also a danger in the case of vehicle wreck if the wheel is spinning freely: the motor is still excited by its magnets and high voltage is present at the motor terminals that can possibly endanger the passengers or rescuers.
- **Magnet demagnetization:** Magnets can be demagnetized by large opposing magnetomotive forces and high temperatures. The critical demagnetization force is different for each magnet material. Great care must be brought to cooling the motor, especially if it is built compact.
- **High-speed capability:** The surface-mounted PM motors cannot reach high speeds because of the limited mechanical strength of the assembly between the rotor yoke and the PMs.
- **Inverter failures in BLDC motor drives:** Because of the PMs on the rotor, BLDC motors present major risks in the case of short-circuit failures of the inverter. Indeed, the rotating rotor is always energized and constantly induces an EMF in the short-circuited windings. A very large current circulates in those windings and an accordingly large torque tends to block the rotor. The dangers of blocking one or several wheels of a vehicle are non-negligible. If the rear wheels are blocked while the front wheels are spinning, the vehicle will spin uncontrollably. If the front wheels are blocked, the driver has no directional control over the vehicle. If only one wheel is blocked, it will induce a yaw torque that will tend to spin the vehicle, which will be difficult to control. In addition to the dangers to the vehicle, it should be noted that the large current resulting from an inverter short circuit poses a risk to demagnetize and destroy the PMs.

Open circuit faults in BLDC motor drives are no direct threat to the vehicle stability. The impossibility of controlling a motor due to an open circuit may, however, pose problems in terms of controlling the vehicle. Because the magnets are always energized and cannot be

controlled, it is difficult to control a BLDC motor in order to minimize the fault. This is a particularly important issue when the BLDC motor is operated in its constant power region. Indeed, in this region, a flux is generated by the stator to oppose the magnet flux and allow the motor to rotate at higher speeds. If the stator flux disappears, the magnet flux will induce a large EMF in the windings, which can be harmful to the electronics or passengers.

BASIC PRINCIPLES OF BLDC MOTOR DRIVES

A BLDC motor drive consists mainly of the BLDC machine, the digital signal processor (DSP) based controller, and the power electronics-based power converter, as shown in Figure 1. Position sensors H1, H2, and H3 sense the position of the machine rotor. The rotor position information is fed to the DSP-based controller, which, in turn, supplies gating signals to the power converter by turning on and turning off the proper stator pole windings of the machine. In this way, the torque and speed of the machines are controlled.

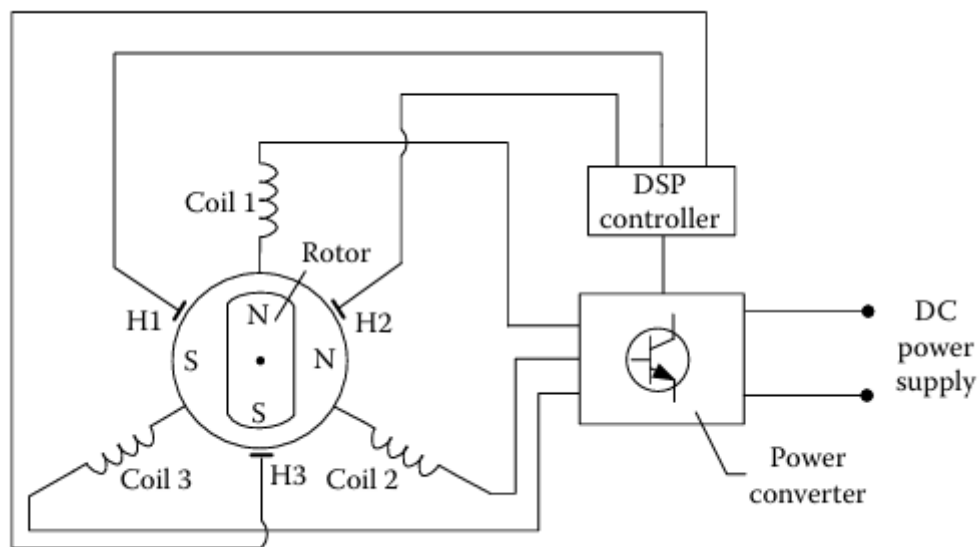


Figure 1: BLDC motor

BLDC MACHINE CONSTRUCTION AND CLASSIFICATION

BLDC machines can be categorized by the position of rotor PM, geometrically, according to the way in which the magnets are mounted on the rotor. The magnets can either be surface mounted or interior mounted. Figure 2a shows the surface-mounted PM rotor. Each PM is mounted on the surface of the rotor. It is easy to build, and specially skewed poles are easily magnetized on this surface-mounted type to minimize cogging torque. But there is a possibility that it will fly apart during high-speed operation.

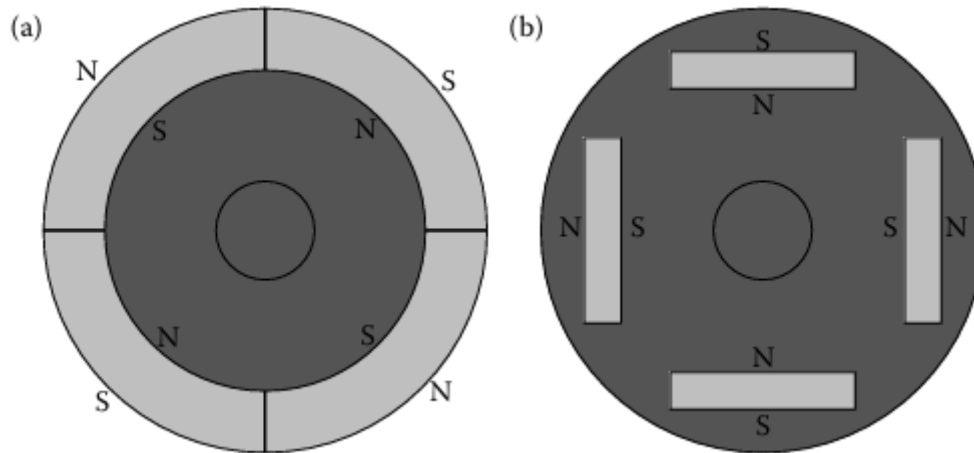


Figure 2: Cross sectional view of the PM rotor: (a) surface-mounted PM rotor; and (b) interior-mounted PM rotor

Figure 2b shows the interior-mounted PM rotor. Each PM is mounted inside the rotor. It is not as common as the surface-mounted type but it is a good candidate for high-speed operation. Note that there is inductance variation for this type of rotor because the PM part is equivalent to air in the magnetic circuit calculation.

In the case of the stator windings, there are two major classes of BLDC motor drives, both of which can be characterized by the shapes of their respective back EMF waveforms, namely trapezoidal and sinusoidal.

The trapezoidal-shaped back EMF BLDC motor is designed to develop trapezoidal back EMF waveforms. It has the following ideal characteristics:

- Rectangular distribution of magnet flux in the air gap.
- Rectangular current waveform.
- Concentrated stator windings.

Excitation waveforms take the form of quasisquare current waveforms with two 60° electrical intervals of zero current excitation per cycle. The nature of the excitation waveforms for trapezoidal back EMF permits some important system simplifications compared to sinusoidal back EMF machines. In particular, the resolution requirements for the rotor position sensor are much lower since only six commutation instants are necessary per electrical cycle. Figure 3 shows the winding configuration of the trapezoidal-shaped back EMF BLDC machine.

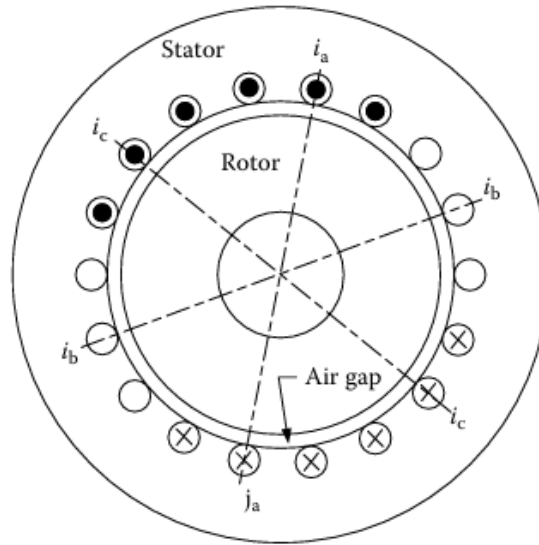


Figure 3: Winding configuration of the trapezoidal-shaped back EMF BLDC.

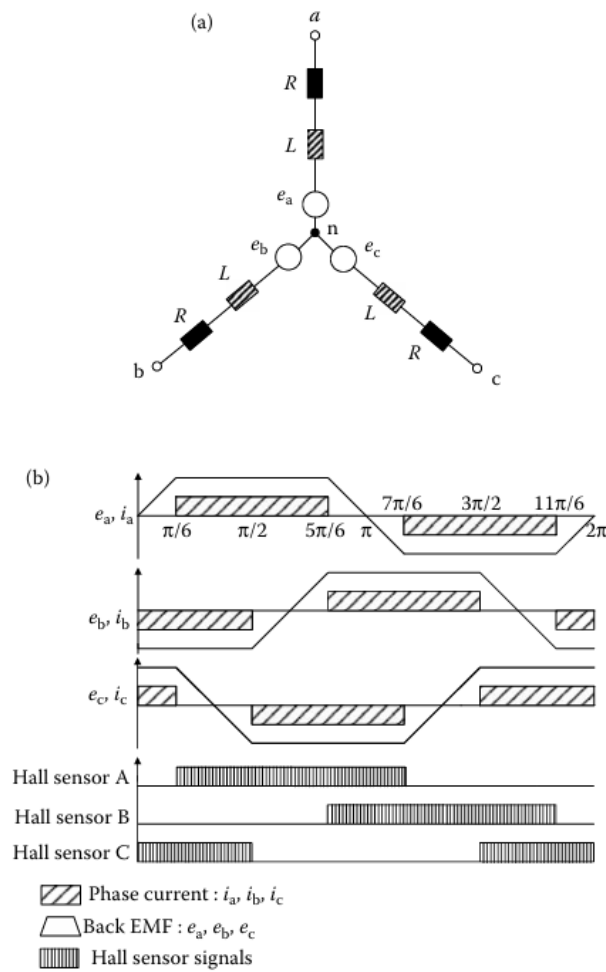


Figure 4: (a) Three-phase equivalent circuit and (b) back EMFs, currents, and Hall sensor signals of a BLDC motor

Figure 4 (a) shows equivalent circuit and (b) shows trapezoidal back EMF, current profiles, and Hall sensor signals of the three-phase BLDC motor drive. The voltages seen in this figure, e_a , e_b , and e_c , are the line-to-neutral back EMF voltages, the result of the PM flux crossing the air gap in a radial direction, and cutting the coils of the stator at a rate proportional to the rotor speed. The coils of the stator are positioned in the standard three-phase full-pitch, concentrated arrangement, and thus the phase trapezoidal back EMF waveforms are displaced by 120° electrical degrees. The current pulse generation is a “ 120° -on and 60° -off” type, meaning each phase current is flowing for $2/3$ of an electrical 360° period, 120° positively and 120° negatively. To drive the motor with maximum and constant torque per ampere, it is desired that the line current pulses be synchronized with the line-neutral back EMF voltages of the particular phase.

A sinusoidal-shaped back EMF BLDC motor is designed to develop sinusoidal back EMF waveforms. It has the following ideal characteristics:

- Sinusoidal distribution of magnet flux in the air gap.
- Sinusoidal current waveforms.
- Sinusoidal distribution of stator conductors.

The most fundamental aspect of the sinusoidal-shaped back EMF motor is that the back EMF generated in each phase winding by the rotation of the magnet should be a sinusoidal wave function of rotor angle. The drive operation of the sinusoidal-shaped back EMF BLDC machine is similar to the AC synchronous motor. It has a rotating stator MMF wave like a synchronous motor and therefore can be analyzed with a phasor diagram. Figure 5 shows the winding configuration of the sinusoidal-shaped back EMF BLDC machine.

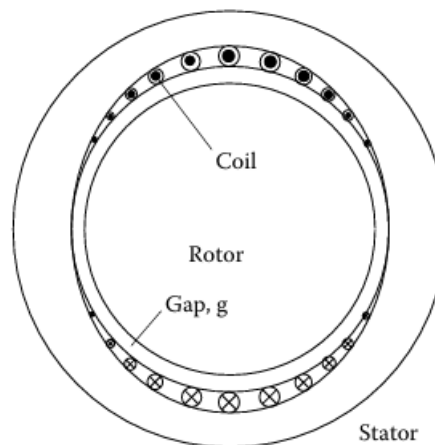


Figure 5: Winding configuration of sinusoidal shaped back emf BLDC

PERFORMANCE ANALYSIS AND CONTROL OF BLDC MACHINES

Speed–torque performance is most important for traction and other applications. As any other electric machine, the torque is produced by the interaction of magnetic field and current. The magnetic field is produced in BLDC by the PM, and the current depends on the source voltage, control, and the back EMF, which is determined by the magnetic field and speed of the machine.

To obtain the desired torque and speed at a given load, the current needs to be controlled.

PERFORMANCE ANALYSIS

The performance analysis of the BLDC machines is based on the following assumption for simplification:

1. The motor is not saturated.
2. Stator resistances of all the windings are equal and self- and mutual inductances are constant.
3. Power semiconductor devices in the inverter are ideal.
4. Iron losses are negligible.

A simplified equivalent circuit of one phase is shown in Figure 6.48, where V_t is the voltage of the power supply, R_s is the resistance of the winding, L_s is the leakage inductance ($L_s = L - M$, where L is the self-inductance of the winding and M is the mutual inductance), and E_s is the back EMF induced in the winding by the rotating rotor.

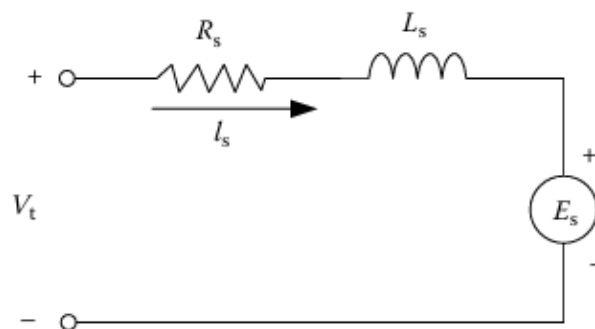


Figure 6: Simplified equivalent circuit of the BLDC motor.

Based on the equivalent circuit of Figure 6.48, the performance of the BLDC motor can be described by

$$V_t = R_s I_s + L_s \frac{dI_s}{dt} + E_s,$$

$$E_s = k_E \omega_r$$

$$T_e = k_T I_s$$

$$T_e = T_L + J \frac{d\omega_r}{dt} + B\omega_r$$

where k_E is the back EMF constant, which is associated with the PMs and rotor structure, ω_r is the angular velocity of the rotor, k_T is the torque constant, T_L is the load torque, and B is the viscous resistance coefficient. For steady-state operation, Equations can be simply reduced to

$$T_e = \frac{(V_t - k_E \omega_r) k_T}{R_s}$$

The speed–torque performance with constant voltage supply is shown in Figure 7.

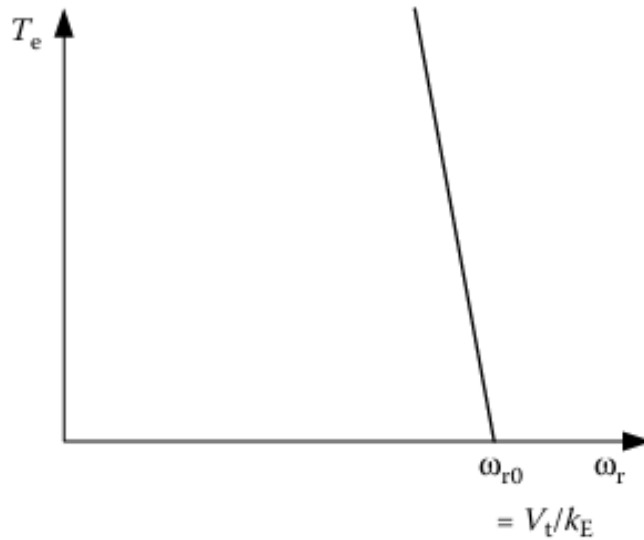


Figure 7: Speed–torque curve in the steady state with constant voltage.

It can be seen from Equation and Figure that at low speed, especially while starting, very high torque is produced, which results in very high current due to the low back EMF. This very high current would damage the stator windings.

With variable voltage supply, the winding current can be restricted to its maximum by actively controlling the voltage; thus, a maximum constant torque can be produced as shown in Figure 8.

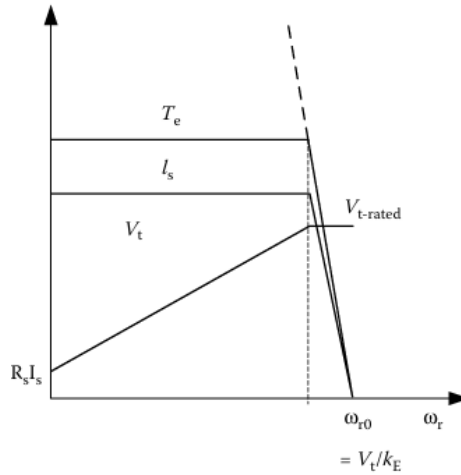


Figure 8: Speed–torque curve in the steady state with variable voltage supply.

For dynamic or transient operation, the performance of the BLDC machine is described by Equations. However, Laplace transform is helpful to simplify the analysis. Equations can be expressed by their Laplace forms as

$$V_t(s) = E_s(s) + (R_s + sL_s)I_s(s)$$

$$E_s(s) = k_E \omega_r(s)$$

$$T_e(s) = k_T I_s(s)$$

$$T_e(s) = T_L(s) + (B + sJ)\omega_r(s)$$

Thus, the transfer function of the BLDC motor drive system is

$$\omega_r(s) = \frac{k_T}{(R_s + sL_s)(sJ + B) + k_T k_E} V_t(s) - \frac{R_s + sL_s}{(R_s + sL_s)(sJ + B) + k_T k_E} T_L(s).$$

L_s and J in Equation represent the electrical and mechanical delay in transient operation. L_s determines how quickly the armature current builds up in response to a step change in the terminal voltage, where the rotor speed is assumed to be constant. J determines how quickly the speed builds up in response to a step change in the terminal voltage.

CONTROL OF BLDC MOTOR DRIVES

In vehicle traction application, the torque produced is required to follow the torque desired by the driver and commanded through the accelerator and brake pedals. Thus, torque control is the basic requirement.

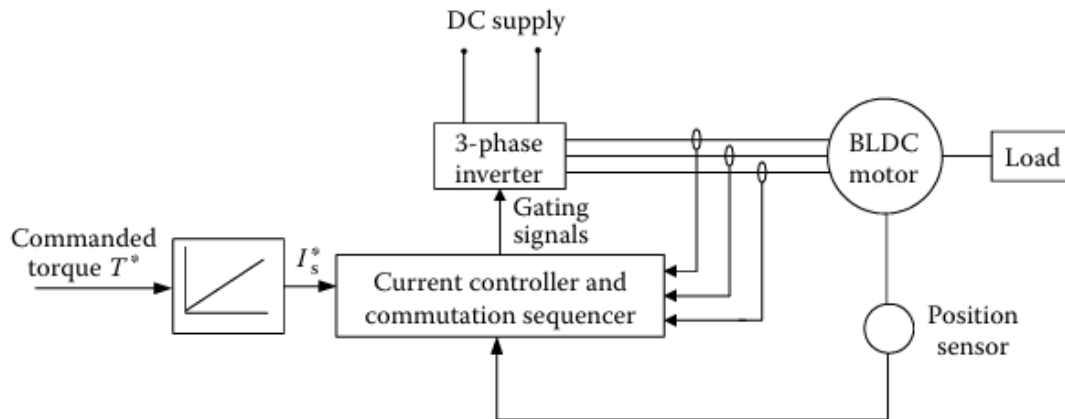


Figure 9: Block diagram of the torque control of the BLDC motor.

Figure 9 shows a block diagram of a torque control scheme for a BLDC motor drive. The desired current I^* is derived from the commanded torque T^* through a torque controller. The current controller and commutation sequencer receive the desired current I^* , position information from the position sensors, and perhaps the current feedback through current transducers, and then produce gating signals. These gating signals are sent to the three-phase inverter (power converter) to produce the desired phase current to the BLDC machine.

In traction application, speed control may be required, cruising control operation for example (see Figure 10). Many high-performance applications include current feedback for torque control. At the minimum, DC bus current feedback is required to protect the drive and machine from over-currents. The controller blocks, “speed controller” may be any type of classical controller such as a proportional-integral (PI) controller or a more advanced controller such as an artificial intelligence control. The “current controller and commutation sequencer” provides the properly sequenced gating signals to the “three-phase inverter” while comparing sensed currents to a reference to maintain a constant peak current control by hysteresis (current chopping) or with a voltage source (PWM)-type current control. Using position information, the commutation sequencer causes the inverter to “electronically commute,” acting as the mechanical commutator of a conventional DC machine. The commutation angle associated with a brushless motor is normally set so that the motor will commute around the peak of the torque angle curve. Considering a three-phase motor, connected in Delta or wye, commutation occurs at electrical angles, which are plus or minus 30° (electrical) from the peaks of the torque-angle curves. When the motor position moves beyond the peaks by an amount equal to

30° (electrical), then the commutation sensors cause the stator phase excitation to switch to move the motor suddenly to -30° relative to the peak of the next torque–angle curve

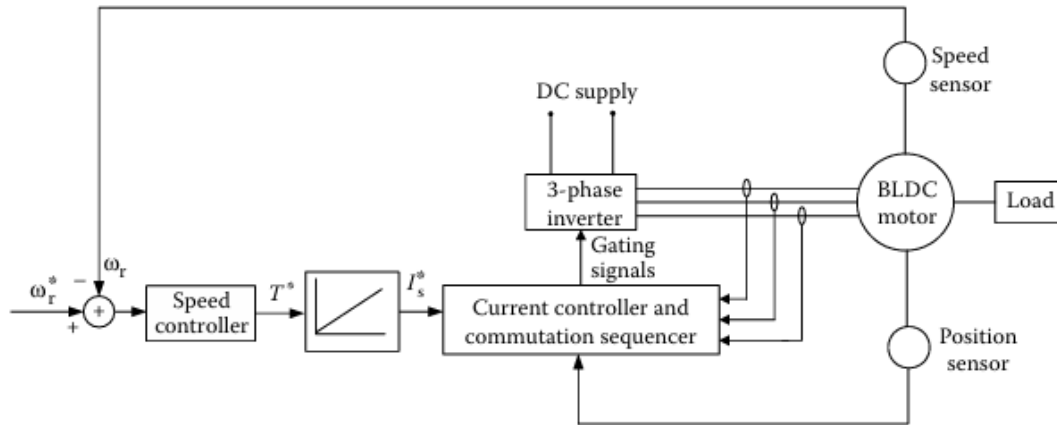


Figure 10: Block diagram of the speed control of the BLDC motor

EXTEND SPEED TECHNOLOGY

As discussed above, PM BLDC machines inherently have a short constant power range due to their rather limited field weakening capability. This is a result of the presence of the PM field, which can only be weakened through production of a stator field component, which opposes the rotor magnetic field. The speed ratio, x , is usually less than 2.

Recently, the use of additional field windings to extend the speed range of PM BLDC motors has been developed. The key is to control the field current in such a way that the air gap field provided by PMs can be weakened during high-speed, constant-power operation. Due to the presence of both PMs and the field windings, these motors are the so-called PM hybrid motors. The PM hybrid motor can achieve a speed ratio of around 4. The optimal efficiency profiles of a PM hybrid motor drive are shown in Figure 11. However, the PM hybrid motors have the drawback of a relatively complex structure. The speed ratio is still not enough to meet the vehicle performance requirement, especially to an off-road vehicle. Thus, a multigear transmission is required.

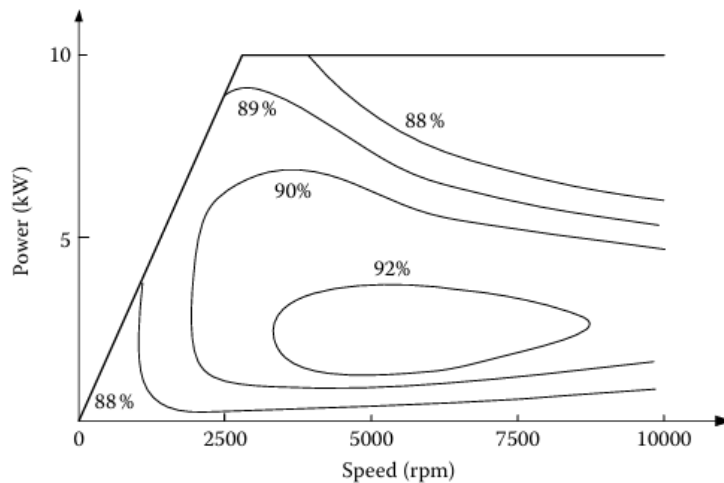


Figure 11: Optimal efficiency profiles of a PM hybrid motor drive

SENSORLESS TECHNIQUES

As mentioned above, the operation of the BLDC motor drives relies mostly on position sensors for obtaining the rotor position information so as to properly perform the turn-on or turn-off of each phase properly. The position sensor is usually either a three-element Hall-effect sensor or an optical encoder. These position sensors are high-cost, fragile elements. Thus, its presence not only enhances the cost of the motor drive but also seriously lowers its reliability and limits its application in some environments, such as the military one.

Position sensorless technology can effectively continue the operation of the system in case the position sensors lose their function. This is crucial in some applications, such as military vehicles. Several sensorless technologies have been developed. The majority of them are based on the voltage, current, and back EMF detection. These techniques can be primarily grouped into four categories:

1. Those using measured currents, voltages, fundamental machine equations, and algebraic manipulations.
2. Those using observers.
3. Those using back EMF methods.
4. Those with novel techniques not falling into the previous three categories.